

Installation of Elasticsearch

- 1) docker pull elastic/elasticsearch:9.3.2
- 2) docker images
- 3) docker network create elastic
- 4) docker run --name elasticsearch --net elastic -p 9200:9200 -e "discovery.type=single-node" -e "xpack.security.enabled=false" -e "xpack.security.http.ssl.enabled=false" -e "ES_JAVA_OPTS=-Xms512m -Xmx512m" -d elastic/elasticsearch:9.3.2
- 5) docker ps
- 6) curl <http://localhost:9200>

Execution of CRUD Operations on Elasticsearch via command prompt

Create an Index

```
curl -X PUT http://localhost:9200/students
```

Insert a Document (Index a Document)

```
curl -X POST http://localhost:9200/students/_doc/1 -H "Content-Type: application/json" -d '{"name": "Vishal", "age": 22, "city": "Hyderabad"}'
```

Get a Document by ID

```
curl http://localhost:9200/students/_doc/1
```

Search all documents:

```
curl http://localhost:9200/students/_search
```

Search by a field value:

```
curl -X GET http://localhost:9200/students/_search -H "Content-Type: application/json" -d '{"query": {"match": {"city": "Hyderabad"}}}'
```

Update a Document

```
curl -X POST http://localhost:9200/students/_update/1 -H "Content-Type: application/json" -d '{"doc": {"age": 23}}'
```

Delete a Document

```
curl -X DELETE http://localhost:9200/students/_doc/1
```

Instead of typing long curl commands, Kibana gives you a browser-based interface to interact with Elasticsearch. Want me to show you how to set up Kibana with your existing Docker setup?

Installation of Kibana

Pull Kibana Image

```
docker pull kibana:9.3.2
```

```
docker run --name kibana --net elastic -p 5601:5601 -d kibana:9.3.2
```

Wait ~1 Minute, Then Open Browser

<http://localhost:5601>

Use Dev Tools Console (Most Important!)

Once inside Kibana:

1. Click the ****≡ hamburger menu**** (top left)
2. Go to ****Management → Dev Tools****
3. You'll see a console like this:

Now Try These in Dev Tools

Create an index

PUT /kmit

Insert a document

```
POST /kmit/_doc/1
{
  "name": "Vishal",
  "age": 22,
  "city": "Hyderabad"
}
```

Search all

GET /kmit/_search

Search by field

```
GET /kmit/_search
{
  "query": {
    "match": {
      "city": "Hyderabad"
    }
  }
}
```

Update

```
POST /kmit/_update/1
{
  "doc": {
    "age": 23
  }
}
```

Delete a document

DELETE /students/_doc/1

In Notes Example:

Run these commands in gitBash

To stop all the containers which are running

```
docker ps -q | xargs -r docker stop
```

To remove all the containers

```
docker ps -a -q | xargs -r docker rm -f
```

To run the elastic search

```
docker run --name elasticsearch --net elastic -p 9200:9200 -e "discovery.type=single-node" -e "xpack.security.enabled=false" -e "xpack.security.http.ssl.enabled=false" -e "ES_JAVA_OPTS=-Xms512m -Xmx512m" -d elastic/elasticsearch:9.3.2
```

To run the Kibana Server

```
docker run --name kibana --net elastic -p 5601:5601 -d kibana:9.3.2
```

creates a movie index:

Put movie

Indexing a document with an autogenerated ID

POST movie/_doc

```
{
  "name": "titanic",
  "release_date": "December 19, 1997",
  "director": "James Cameron",
  "description": "Seventeen-year-old Rose hails from an aristocratic family and is set to be married. When she boards the Titanic, she meets Jack Dawson, an artist, and falls in love with him."
}
```

Indexing a document with a specified ID

Post movie/_doc/1

```
{
  "name": "Titanic",
  "release_date": "December 19, 1997",
  "director": "James Cameron",
  "description": "Seventeen-year-old Rose hails from an aristocratic family and is set to be married. When she boards the Titanic, she meets Jack Dawson, an artist, and falls in love with him."
}
```

The _create endpoint

POST movie/_create/1

```
{
  "name": "avatar",
  "release_date": "December 17, 2009",
  "director": "James Cameron"
}
```

Reading a document

GET movie/_doc/1

Updating a document

```
POST movie/_update/1
{
  "doc": {
    "release_date": "December 02, 2022",
    "director": "Donald Trump"
  }
}
```

Deleting a document

```
DELETE movie/_doc/1
```

Mapping

Dynamic mapping

PUT article

GET article/_mapping

POST article/_create/1

```
{
  "title": "How to Use Elasticsearch",
  "number_of_authors": 1,
  "is_published": true,
  "publish_date": "2022-01-01",
  "tags": ["elasticsearch", "search", "nosql"],
  "body": "Elasticsearch is a powerful search engine that makes it easy to search"
}
```

GET article/_mapping

Explicit Mapping

PUT music

```
{
  "mappings": {
    "properties": {
      "title": {
        "type": "text"
      },
      "author": {
        "type": "text"
      },
      "publish_date": {
        "type": "date"
      }
    }
  }
}
```

```
    "tags": {
      "type": "keyword"
    }
  }
}
```

GET music/_mapping

```
PUT music/_mapping
{
  "properties": {
    "rate": {
      "type": "double"
    },
    "number_of_listeners": {
      "type": "integer"
    }
  }
}
```

GET music/_mapping

Full-Text Search: Match Query

```
PUT book
{
  "mappings": {
    "properties": {
      "title": {
        "type": "text",
        "analyzer": "english"
      },
      "author": {
        "type": "keyword"
      },
      "publisher": {
        "type": "keyword"
      },
      "year": {
        "type": "integer"
      }
    }
  }
}
```

```
PUT /book/_create/1
{
  "title": "The Indexing Companion",
  "author": "Glenda Browne",
  "publisher": "Information Today, Inc.",
  "year": 2007
}
```

```
}

PUT /book/_create/2
{
  "title": "Indexing Books",
  "author": "Nancy C. Mulvany",
  "publisher": "University of Chicago Press",
  "year": 1994
}
```

```
PUT /book/_create/3
{
  "title": "The Catcher in the Rye",
  "author": "J.D. Salinger",
  "publisher": "Brown and Company",
  "year": 1951
}
```

```
PUT /book/_create/4
{
  "title": "The Complete Book of Indexing",
  "author": "Nancy C. Mulvany",
  "publisher": "Information Today, Inc.",
  "year": 1991
}
```

```
GET book/_search
{
  "query": {
    "match": {
      "title": "Indexing books"
    }
  }
}
```

Full-Text Search: Match Query Parameters

```
PUT book
{
  "mappings": {
    "properties": {
      "title": {
        "type": "text",
        "analyzer": "english"
      },
      "description": {
        "type": "text",
        "analyzer": "english"
      }
    }
  }
}
```

```
PUT /book/_create/1
{
  "title": "Art History: A Very Short Introduction",

```

```
    "description": "A brief yet informative overview of art history, presenting key concepts, themes, and developments in art across different civilizations and time periods."
  }
}
```

```
PUT /book/_create/2
```

```
{
  "title": "Art History: The Basics",
  "description": "A concise guide that provides a foundational understanding of art history, discussing key movements, major artists, and significant artworks."
}
```

```
PUT /book/_create/3
```

```
{
  "title": "Art History: A Critical Introduction",
  "description": "An in-depth exploration of art history as a discipline, examining its methodologies, theories, and the social and cultural contexts that shape artistic expression."
}
```

```
PUT /book/_create/4
```

```
{
  "title": "The Power of Art",
  "description": "A captivating exploration of the transformative power of art, delving into the emotional, cultural, and societal impact of artistic creations across different mediums and genres."
}
```

```
GET /book/_search
```

```
{
  "query": {
    "match": {
      "title": {
        "query": "art history"
      }
    }
  }
}
```

```
GET /book/_search
```

```
{
  "query": {
    "match": {
      "title": {
        "query": "art history",
        "operator": "and"
      }
    }
  }
}
```

```
GET /book/_search
```

```
{
  "query": {
    "match": {
      "description": {
        "query": "art history and its methodologies",
        "minimum_should_match": "75%"
      }
    }
  }
}
```

```
}  
}  
}
```

Term-Level Queries: Term Query

```
PUT /product3
```

```
{  
  "mappings": {  
    "properties": {  
      "name": {  
        "type": "text"  
      },  
      "category": {  
        "type": "keyword"  
      },  
      "price": {  
        "type": "float"  
      }  
    }  
  }  
}
```

```
PUT /product3/_create/1
```

```
{  
  "name": "iPhone 12",  
  "category": "electronics_mobile",  
  "price": 999  
}
```

```
PUT /product3/_create/2
```

```
{  
  "name": "Samsung Galaxy S20",  
  "category": "electronics_mobile",  
  "price": 899  
}
```

```
PUT /product3/_create/3
```

```
{  
  "name": "Sony Bravia 4K TV",  
  "category": "electronics_monitor",  
  "price": 1499  
}
```

```
PUT /product3/_create/4
```

```
{  
  "name": "Dell XPS 13",  
  "category": "electronics_laptop",  
  "price": 1499  
}
```



```
}

GET product3/_search
{
  "query": {
    "term": {
      "category": "electronics_mobile"
    }
  }
}
```

Term-Level Queries: Range Query

```
PUT /product2
{
  "mappings": {
    "properties": {
      "name": {
        "type": "text"
      },
      "category": {
        "type": "keyword"
      },
      "price": {
        "type": "float"
      }
    }
  }
}

PUT /product2/_create/1
{
  "name": "iPhone 12",
  "category": "electronics_mobile",
  "price": 1300
}

PUT /product2/_create/2
{
  "name": "Samsung Galaxy S20",
  "category": "electronics_mobile",
  "price": 1500
}

PUT /product2/_create/3
{
  "name": "Sony Bravia 4K TV",
```

```

    "category": "electronics_monitor",
    "price": 900
}

PUT /product2/_create/4
{
  "name": "Dell XPS 13",
  "category": "electronics_laptop",
  "price": 1350
}

PUT /product2/_create/5
{
  "name": "Samsung 85 Inch Smart Crystal UHD 4K",
  "category": "electronics_monitor",
  "price": 1200
}

PUT /product2/_create/6
{
  "name": "Dell LATITUDE 7000 13.3",
  "category": "electronics_laptop",
  "price": 1400
}

GET product2/_search
{
  "query": {
    "range": {
      "price": {
        "gte": 1200,
        "lte": 1500
      }
    }
  }
}

```

Compound Queries: Boolean Query

```

// Creating data

PUT /product1
{
  "mappings": {
    "properties": {
      "name": {
        "type": "text"
      }
    }
  }
}

```

```
    },
    "category": {
      "type": "keyword"
    },
    "price": {
      "type": "float"
    },
    "quantity": {
      "type": "integer"
    },
    "availability": {
      "type": "keyword"
    }
  }
}
```

PUT /product1/_create/1

```
{
  "name": "iPhone 12",
  "category": "Electronics",
  "price": 1099.99,
  "quantity": 50,
  "availability": "in stock"
}
```

PUT /product1/_create/2

```
{
  "name": "Samsung Galaxy S21",
  "category": "Electronics",
  "price": 899.99,
  "quantity": 25,
  "availability": "in stock"
}
```

PUT /product1/_create/3

```
{
  "name": "Sony PlayStation 5",
  "category": "Electronics",
  "price": 499.99,
  "quantity": 10,
  "availability": "out of stock"
}
```

PUT /product1/_create/4

```
{
  "name": "KitchenAid Stand Mixer",
  "category": "Home & Kitchen",

```

```
"price": 349.99,
"quantity": 15,
"availability": "in stock"
}

PUT /product1/_create/5
{
  "name": "Nike Air Max 90",
  "category": "Fashion",
  "price": 129.99,
  "quantity": 30,
  "availability": "in stock"
}

PUT /product1/_create/6
{
  "name": "Amazon Echo Dot",
  "category": "Electronics",
  "price": 49.99,
  "quantity": 5,
  "availability": "in stock"
}

// Queries

GET product1/_search
{
  "query": {
    "bool": {
      "must": [
        {
          "term": {
            "category": "Electronics"
          }
        },
        {
          "range": {
            "price": {
              "lt": 500
            }
          }
        }
      ]
    }
  }
}

GET product1/_search
```

```
{
  "query": {
    "bool": {
      "must": [
        {
          "term": {
            "availability": "in stock"
          }
        }
      ],
      "must_not": [
        {
          "term": {
            "category": "Electronics"
          }
        }
      ]
    }
  }
}
```

GET product1/_search

```
{
  "query": {
    "bool": {
      "must": [
        {
          "range": {
            "price": {
              "gte": 100,
              "lte": 1000
            }
          }
        }
      ],
      "must_not": [
        {
          "term": {
            "category": "Fashion"
          }
        }
      ],
      "should": [
        {
          "match": {
            "name": "Samsung"
          }
        }
      ]
    }
  }
}
```

```
}
  }
}
}
```

Aggregations – Bucket Aggregation

Dataset to be used – [click on the link to download](#)

PUT product

```
{
  "mappings": {
    "properties": {
      "name": {
        "type": "text"
      },
      "description": {
        "type": "text"
      },
      "price": {
        "type": "double"
      },
      "category": {
        "type": "keyword"
      },
      "brand": {
        "type": "keyword"
      },
      "color": {
        "type": "keyword"
      },
      "availability": {
        "type": "keyword"
      },
      "rating": {
        "type": "float"
      },
      "seller": {
        "type": "keyword"
      }
    }
  }
}
```

POST product/_bulk

```
{"index":{"_id":1}}{"name":"Product1","description":"Smartelectronicdevice",
price":510,"category":"laptop","brand":"Samsung","color":"white","availabilit
y":"out_of_stock","rating":3.6,"seller":"Flipkart"}{"index":{"_id":2}}{"name"
:"Product2","description":"Advancedgadget","price":520,"category":"tablet","b
rand":"Dell","color":"blue","availability":"in_stock","rating":3.7,"seller":"
```

```
Croma"}{"index":{"_id":3}}{"name":"Product3","description":"Smartelectronicdevice","price":530,"category":"mobile","brand":"HP","color":"gray","availability":"out_of_stock","rating":3.8,"seller":"Amazon"}{"index":{"_id":4}}{"name":"Product4","description":"Advancedgadget","price":540,"category":"laptop","brand":"Lenovo","color":"silver","availability":"in_stock","rating":3.9,"seller":"Flipkart"}{"index":{"_id":5}}{"name":"Product5","description":"Smartelectronicdevice","price":550,"category":"tablet","brand":"Asus","color":"green","availability":"out_of_stock","rating":4.0,"seller":"Croma"}{"index":{"_id":6}}{"name":"Product6","description":"Advancedgadget","price":560,"category":"mobile","brand":"Acer","color":"gold","availability":"in_stock","rating":4.1,"seller":"Amazon"}{"index":{"_id":7}}{"name":"Product7","description":"Smartelectronicdevice","price":570,"category":"laptop","brand":"Realme","color":"black","availability":"out_of_stock","rating":4.2,"seller":"Flipkart"}{"index":{"_id":8}}{"name":"Product8","description":"Advancedgadget","price":580,"category":"tablet","brand":"OnePlus","color":"white","availability":"in_stock","rating":4.3,"seller":"Croma"}{"index":{"_id":9}}{"name":"Product9","description":"Smartelectronicdevice","price":590,"category":"mobile","brand":"Google","color":"blue","availability":"out_of_stock","rating":4.4,"seller":"Amazon"}{"index":{"_id":10}}{"name":"Product10","description":"Advancedgadget","price":600,"category":"laptop","brand":"Apple","color":"gray","availability":"in_stock","rating":4.5,"seller":"Flipkart"}{"index":{"_id":11}}{"name":"Product11","description":"Smartelectronicdevice","price":610,"category":"tablet","brand":"Samsung","color":"silver","availability":"out_of_stock","rating":4.6,"seller":"Croma"}{"index":{"_id":12}}{"name":"Product12","description":"Advancedgadget","price":620,"category":"mobile","brand":"Dell","color":"green","availability":"in_stock","rating":4.7,"seller":"Amazon"}{"index":{"_id":13}}{"name":"Product13","description":"Smartelectronicdevice","price":630,"category":"laptop","brand":"HP","color":"gold","availability":"out_of_stock","rating":4.8,"seller":"Flipkart"}{"index":{"_id":14}}{"name":"Product14","description":"Advancedgadget","price":640,"category":"tablet","brand":"Lenovo","color":"black","availability":"in_stock","rating":4.9,"seller":"Croma"}{"index":{"_id":15}}{"name":"Product15","description":"Smartelectronicdevice","price":650,"category":"mobile","brand":"Asus","color":"white","availability":"out_of_stock","rating":3.5,"seller":"Amazon"}{"index":{"_id":16}}{"name":"Product16","description":"Advancedgadget","price":660,"category":"laptop","brand":"Acer","color":"blue","availability":"in_stock","rating":3.6,"seller":"Flipkart"}{"index":{"_id":17}}{"name":"Product17","description":"Smartelectronicdevice","price":670,"category":"tablet","brand":"Realme","color":"gray","availability":"out_of_stock","rating":3.7,"seller":"Croma"}{"index":{"_id":18}}{"name":"Product18","description":"Advancedgadget","price":680,"category":"mobile","brand":"OnePlus","color":"silver","availability":"in_stock","rating":3.8,"seller":"Amazon"}{"index":{"_id":19}}{"name":"Product19","description":"Smartelectronicdevice","price":690,"category":"laptop","brand":"Google","color":"green","availability":"out_of_stock","rating":3.9,"seller":"Flipkart"}{"index":{"_id":20}}{"name":"Product20","description":"Advancedgadget","price":700,"category":"tablet","brand":"Apple","color":"gold","availability":"in_stock","rating":4.0,"seller":"Croma"}{"index":{"_id":21}}{"name":"Product21","description":"Smartelectronicdevice","price":710,"category":"mobile","brand":"Samsung","color":"black","a
```

```
availability":"out_of_stock","rating":4.1,"seller":"Amazon"}{"index":{"_id":22}}{"name":"Product22","description":"Advancedgadget","price":720,"category":"laptop","brand":"Dell","color":"white","availability":"in_stock","rating":4.2,"seller":"Flipkart"}{"index":{"_id":23}}{"name":"Product23","description":"Smartelectronicdevice","price":730,"category":"tablet","brand":"HP","color":"blue","availability":"out_of_stock","rating":4.3,"seller":"Croma"}{"index":{"_id":24}}{"name":"Product24","description":"Advancedgadget","price":740,"category":"mobile","brand":"Lenovo","color":"gray","availability":"in_stock","rating":4.4,"seller":"Amazon"}{"index":{"_id":25}}{"name":"Product25","description":"Smartelectronicdevice","price":750,"category":"laptop","brand":"Asus","color":"silver","availability":"out_of_stock","rating":4.5,"seller":"Flipkart"}{"index":{"_id":26}}{"name":"Product26","description":"Advancedgadget","price":760,"category":"tablet","brand":"Acer","color":"green","availability":"in_stock","rating":4.6,"seller":"Croma"}{"index":{"_id":27}}{"name":"Product27","description":"Smartelectronicdevice","price":770,"category":"mobile","brand":"Realme","color":"gold","availability":"out_of_stock","rating":4.7,"seller":"Amazon"}{"index":{"_id":28}}{"name":"Product28","description":"Advancedgadget","price":780,"category":"laptop","brand":"OnePlus","color":"black","availability":"in_stock","rating":4.8,"seller":"Flipkart"}{"index":{"_id":29}}{"name":"Product29","description":"Smartelectronicdevice","price":790,"category":"tablet","brand":"Google","color":"white","availability":"out_of_stock","rating":4.9,"seller":"Croma"}{"index":{"_id":30}}{"name":"Product30","description":"Advancedgadget","price":800,"category":"mobile","brand":"Apple","color":"blue","availability":"in_stock","rating":3.5,"seller":"Amazon"}{"index":{"_id":31}}{"name":"Product31","description":"Smartelectronicdevice","price":810,"category":"laptop","brand":"Samsung","color":"gray","availability":"out_of_stock","rating":3.6,"seller":"Flipkart"}{"index":{"_id":32}}{"name":"Product32","description":"Advancedgadget","price":820,"category":"tablet","brand":"Dell","color":"silver","availability":"in_stock","rating":3.7,"seller":"Croma"}{"index":{"_id":33}}{"name":"Product33","description":"Smartelectronicdevice","price":830,"category":"mobile","brand":"HP","color":"green","availability":"out_of_stock","rating":3.8,"seller":"Amazon"}{"index":{"_id":34}}{"name":"Product34","description":"Advancedgadget","price":840,"category":"laptop","brand":"Lenovo","color":"gold","availability":"in_stock","rating":3.9,"seller":"Flipkart"}{"index":{"_id":35}}{"name":"Product35","description":"Smartelectronicdevice","price":850,"category":"tablet","brand":"Asus","color":"black","availability":"out_of_stock","rating":4.0,"seller":"Croma"}{"index":{"_id":36}}{"name":"Product36","description":"Advancedgadget","price":860,"category":"mobile","brand":"Acer","color":"white","availability":"in_stock","rating":4.1,"seller":"Amazon"}{"index":{"_id":37}}{"name":"Product37","description":"Smartelectronicdevice","price":870,"category":"laptop","brand":"Realme","color":"blue","availability":"out_of_stock","rating":4.2,"seller":"Flipkart"}{"index":{"_id":38}}{"name":"Product38","description":"Advancedgadget","price":880,"category":"tablet","brand":"OnePlus","color":"gray","availability":"in_stock","rating":4.3,"seller":"Croma"}{"index":{"_id":39}}{"name":"Product39","description":"Smartelectronicdevice","price":890,"category":"mobile","brand":"Google","color":"silver","availability":"out_of_stock","rating":4.4,"seller":"Amazon"}{"index":{"_id":40}}{"name":"Product40","description":"Advancedgadget","price":90
```



```
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Aggregations – Metric Aggregation

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GET product/_search
{
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GET /product/_search

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